



EAST WEST UNIVERSITY

LAB REPORT

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Course Name: Electronic Circuits

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Group No: 10A

Submitted By:

Name	Student ID
Naheda Islam Ela	2024-3-60-080
Abdullah Al Junaid	2024-3-60-082
Jumana Haider	2024-3-60-594

Submitted To:

Dr. Md. Habibur Rahman

Adjunct faculty

Department of Computer Science and Engineering

East west University

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Experiment No.1

Title- To Study the I-V Characteristics of a P-N Junction Diode

Objective:

- To analyze the current-voltage (I-V) characteristics of a p-n junction diode under forward and reverse bias conditions.
- To study the behavior of current due to a semiconductor.
- Learn about resistors and diode.
- To know how to build temporary circuit on breadboard
- To understand its rectifying behavior.
- To find out about how to measure V_d and I_d .

Theory:

A P-N junction diode is formed by joining a p-type semiconductor and an n-type semiconductor. It is one of the most important electronic components because it allows current to flow mainly in one direction.

Forward Bias

When the positive terminal of a power supply is connected to the p-side and the negative terminal is connected to the n-side, the diode is said to be forward biased. In this condition, the diode conducts current easily. As the applied voltage increases, the current through the diode also increases rapidly after a certain voltage is reached.

Reverse Bias

When the connections are reversed (positive terminal to the n-side and negative terminal to the p-side), the diode is reverse biased. In this condition, the diode blocks the flow of current. Only a very small leakage current, usually in the microampere (μA) range, flows through the diode.

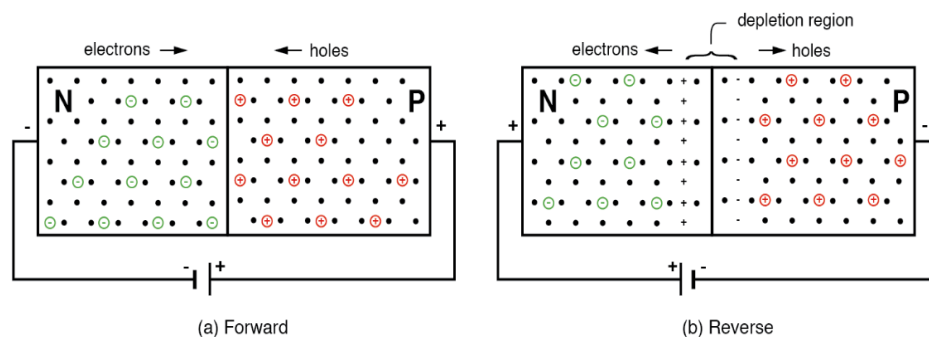


Figure 1: PN Diode in Forward and Reverse Bias

If the reverse voltage is increased beyond a certain limit, the diode undergoes breakdown, causing a large reverse current to flow. This condition can permanently damage a normal diode. Therefore, breakdown testing is not performed in this experiment.

In this experiment, different voltages are applied across the diode and the corresponding current values are measured. By plotting diode current (I_D) against diode voltage (V_D), the I-V characteristics of the diode can be observed and analyzed. This graph shows how the diode behaves under different voltage conditions and helps us understand its operating characteristics.

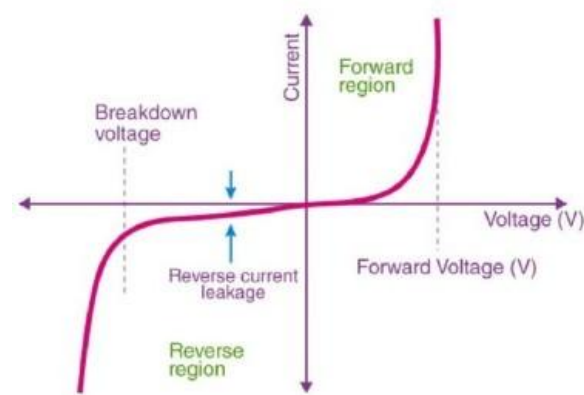


Figure 2: I-V Characteristic of diode

Apparatus Required:

1. Diode -1 piece
2. Resistor (2.2K)
3. Breadboard
4. Connecting wires
5. Multimeter
6. DC power supply
7. Ammeter

Working Diagram:

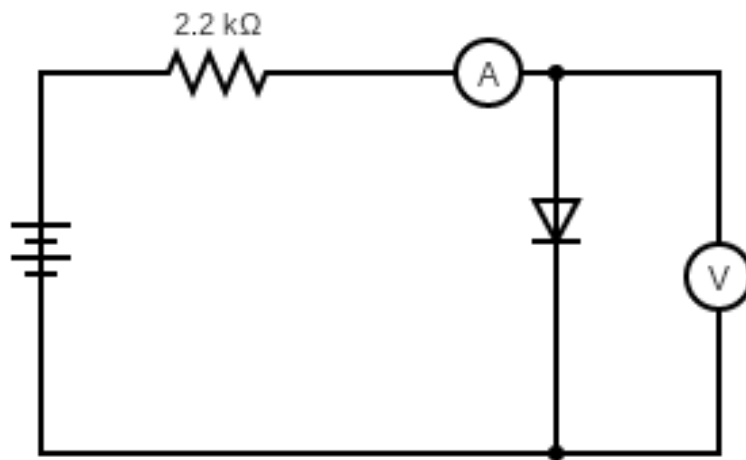


Figure 3: PN Diode in Forward Bias

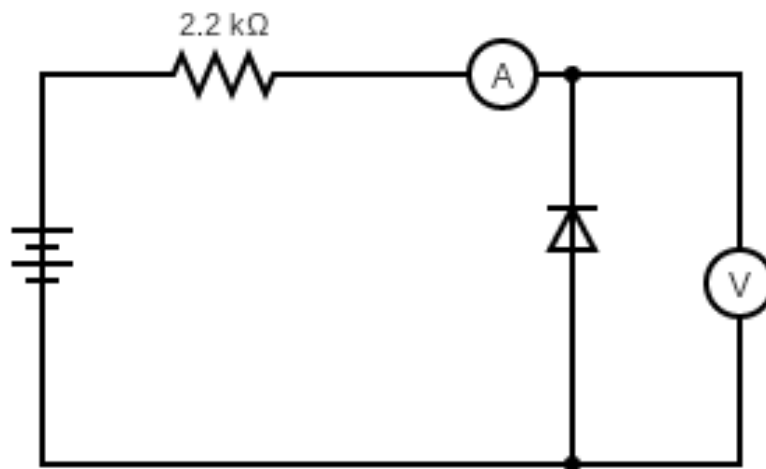


Figure 4: PN Diode in Reverse Bias

Data sheets:

Forward Bias:

No. of Observation	V_D (Volt)	I_D (mA)	V_S (Volt)
1	0.2	0	0.3
2	0.4	0	0.5
3	0.5	0.2	1
4	0.55	0.6	2
5	0.6	1.8	4.6
6	0.61	2.4	5.9
7	0.62	2.8	6.9
8	0.63	3.4	8
9	0.65	5.3	12.1
10	0.66	6.8	15.4
11	0.67	8.5	18.9
12	0.68	10	21.8

Reverse Bias:

No. of Observation	V_D (Volt)	I_D (mA)	V_S (Volt)
1	0	0	0
2	5	0	5.1
3	10	0	10.1
4	15	0	15.2
5	20	0	20.2
6	25	0	25.1
7	30	0	30

Graph:

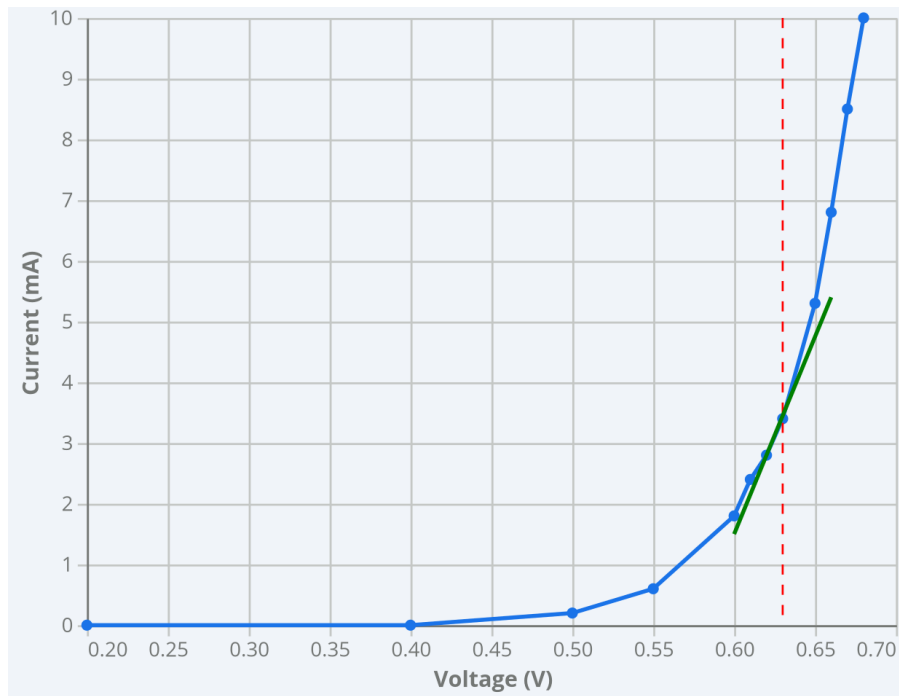


Figure 5: Diode in Forward Bias



Figure 6: Diode in Reverse Bias

Result

The I V characteristics of the P N junction diode were observed in both forward and reverse bias conditions. In forward bias, almost no current flowed below 0.63 V. After about 0.63 V, the current increased rapidly with a small increase in voltage, indicating that the diode started conducting. Therefore, the knee voltage of the diode is approximately 0.63 V.

In reverse bias, no measurable current was observed up to 30 V. This shows that the diode blocked the flow of current in the reverse direction and behaved as expected.

Discussion

The experimental results closely matched the expected behavior of a P–N junction diode. In forward bias, very little current flowed initially, but once the applied voltage exceeded the knee voltage, the current increased rapidly. This confirmed that the diode offers low resistance in the forward direction and allows current to pass easily after overcoming the barrier potential.

In reverse bias, the diode blocked the flow of current and behaved like an insulator, allowing only a negligible reverse current. Thus, the experiment verified the unidirectional conduction property of a P–N junction diode and its ability to act as a rectifier.

Some differences between the theoretical and experimental values may have occurred due to limitations of the measuring instruments, fluctuations in the power supply, and errors while taking readings. Temperature variations and contact resistance in the circuit may also have affected the results. Taking more data points and using more precise measuring instruments could improve the accuracy and produce a smoother I–V characteristic curve.

Overall, the experiment successfully demonstrated the I–V characteristics of a P–N junction diode and provided a better understanding of its operation and practical applications in electronic circuits.